

4. There are 33 known isotopes of krypton. One of the isotopes – ^{81}Kr – decays by electron capture.

(a) Write an equation to show this decay. [1]

- (b) Krypton-81 has been used for dating old groundwater. It takes 6.87×10^5 years for 2.0 g of ^{81}Kr to decay to 0.25 g. Calculate its half-life. [1]

Half-life = years

5. Boron has a relative atomic mass of 10.8. It has only two naturally-occurring isotopes, one of which has an abundance of 80.0% and a mass of 11. Calculate the mass of the second isotope. [2]

Mass =



4. Phosphorus(V) chloride reacts with water to form phosphoric acid and hydrogen chloride. The reactants and products are shown in the equation below but the equation is **not** balanced.



- (a) Balance the equation. [1]
- (b) Calculate the atom economy for the formation of phosphoric acid in this reaction. [2]

Atom economy = %

5. (a) Give the meaning of the term *electronegativity*. [1]

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- (b) Explain why electronegativity increases across a period in the Periodic Table. [1]

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6. Bromine is produced commercially from the bromide ions in sea water by reaction with chlorine.

- (a) Give the **ionic** equation for this reaction. [1]

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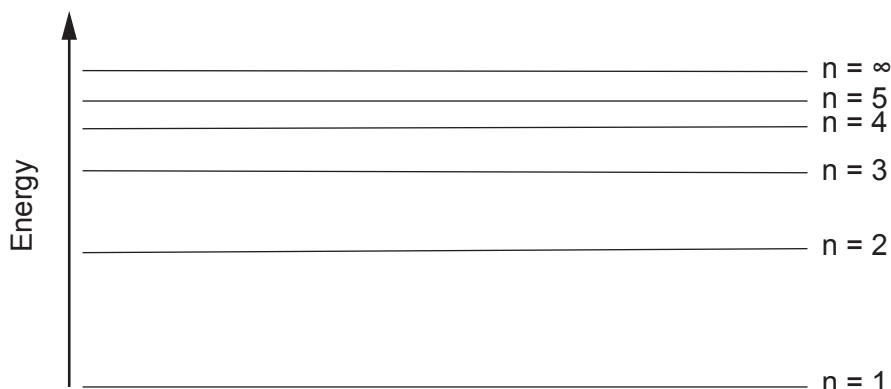
- (b) Describe a test, apart from the use of chlorine, to show that a solution contains bromide ions. Give the reagent(s) and observation(s). [1]

Reagent(s)

Observation(s)



4. The diagram below shows the electron energy levels for a hydrogen atom.



- (a) On the diagram, draw an arrow to represent the transition corresponding to the ionisation of the atom. Label this arrow **A**. [1]
- (b) On the diagram, draw an arrow to represent the transition corresponding to the first line in the visible region in the atomic spectrum. Label this arrow **B**. [1]

5. A student said that $^{32}_{16}\text{S}^{2-}$ and $^{40}_{20}\text{Ca}^{2+}$ have the same electronic configuration.

Do you agree? Give a reason for your answer. [1]

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6. Copper can be extracted from copper(II) oxide using hydrogen.

If the atom economy for this reaction is 78% and 4.2 g of copper is formed, calculate the total mass of the reactants. [2]

Mass of reactants = g

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